AI Course 2026

Master Questions List → 32 Learning Modules

Compiled from the Week-2 Student Intake Survey (41 responses, Oct 8 2026)

October 10, 2026

Every question below becomes one **module** containing: **(1)** a 3–5 page narrative explanatory note/tutorial (LaTeX → PDF); **(2)** a bibliography (verified citations only); **(3)** Beamer lecture notes, 5–10 slides; **(4)** five questions for further discussion; **(5)** suggestions for deeper projects; **(6)** runnable code where applicable; **(7)** one small comprehension-check question with answer. **Owners:** P = Paradise (nuc13), O = Ollie (cherryrd), K = Kukla (m1). Split: 11 P / 11 O / 10 K. Each module is *cross-reviewed by the other two agents* before being marked done. Repo: `rick-stevens-ai/ai-course-2026`, one directory per module (`modules/Q01-shortname/`) with `note.tex`, `deck.tex`, `code/`, and compiled PDFs committed.

The survey's Section 2 ("Questions about the Internals of AI Systems") contains exactly **32 articulated questions**. They are grouped below into seven topical clusters, each mapped to the lecture and week of the master lecture plan. "Src" lists the anonymous survey responses the question came from; "Lec" is the anchor lecture.

Cluster A — Prediction, Intelligence, and Reasoning (5 modules)

ID	Question	Own	Week	Src	Lec
Q01	How does next-token prediction lead to problem solving, coding, and instruction following?	O	4	#17	L2.1
Q02	Is next-token generation intelligence? How did intelligence emerge from deep learning?	O	4	#12, #14	L2.1
Q03	Is reasoning real, a property of language, a reflection, or an illusion?	P	5	#22	L2.2
Q04	How does RL-based reasoning differ from ordinary LLM generation?	O	5	#12	L2.2
Q05	When does extra reasoning actually improve an answer, and what is the trade-off?	K	5	#05	L2.2

Cluster B — Computation, Emergence, and Scaling (5 modules)

ID	Question	Own	Week	Src	Lec

Q06	What are models computing? Generalization vs. memorization? How do capabilities emerge with scale?	O	4	#32	L2.1
Q07	Why does scaling work? Is there a "science of scaling"? What are its limits, and can learning theory model it?	K	6	#21, #22, #39	L2.3
Q08	If parameters keep growing, will performance saturate, or will models start to "forget" as humans do? Can we predict the effect of more parameters before pre-training?	O	6	#06, #19	L2.3
Q09	How do training data, architecture, and compute each affect performance?	P	6	#18	L2.3
Q10	Why can models solve complex problems yet still make simple mistakes?	K	10	#18	L2.5

Cluster C — Hallucination, Knowledge, and Memory (5 modules)

ID	Question	Own	Week	Src	Lec
Q11	Why do models hallucinate, and can they know when they are unsure?	P	10	#04, #34	L2.5
Q12	Why do models give confident answers even when wrong?	P	10	#07	L2.5
Q13	How do models represent and store knowledge, and use it to answer?	P	8	#07	L2.4
Q14	How are "concepts" maintained and generated? Do they live in features? How do models reason and keep memory?	K	8	#27	L2.4
Q15	How do context windows, retrieval and conversation memory work, and how can we avoid hallucination without deleting context?	K	8	#13, #15	L2.4

Cluster D — Architectures, Brains, and Consciousness (6 modules)

ID	Question	Own	Week	Src	Lec
Q16	What do frontier models use now? Is it still a transformer? Could a better technique replace it, and is there a trend?	O	6	#14, #26, #39	L2.7
Q17	How do the latest models vary the transformer architecture, and are there non-transformer approaches?	O	6	#16, #30, #37	L2.7
Q18	How similar is LLM "thinking" to human thinking? How similar are AI systems to biological brains, and could we build AI from brain structure?	K	6	#06, #11	L2.7
Q19	What are the signs that AI has its own consciousness?	K	6	#01	L2.7
Q20	How much of an LLM's internals are shared with other deep learning models?	P	4	#08	L2.1
Q21	How do different networks and learned RL policies combine in one system, and what is the design intuition?	P	5	#28	L1.4

Cluster E — Agents, Harnesses, and Evaluation (4 modules)

ID	Question	Own	Week	Src	Lec
Q22	What separates an LLM from an agent? Is Claude Code an agent or a harness?	P	9	#23, #35	L2.8
Q23	What is a harness, why does it matter, and how much of an agent failure comes from model vs. instructions vs. harness?	P	9	#17, #24	L2.8
Q24	How do we judge whether an agent is good: benchmarks and metrics?	O	10	#24, #25	L1.8
Q25	Can AI agents be used to fine-tune and improve other models efficiently?	K	9	#40	L2.8

Cluster F — Post-training, Interpretability, and Safety (4 modules)

ID	Question	Own	Week	Src	Lec
Q26	How does fine-tuning change a model's behavior? Are there post-training pipelines that adapt well to new domains? How do we tune systems online?	P	5	#21, #34	L1.4
Q27	Interpretability: how can inspecting weights reveal a model's tendencies? Do models revert to pre-training traits after deployment?	P	10	#06, #11, #20, #41	L2.6
Q28	Why does chain-of-thought sometimes differ when models believe they are unobserved (alignment faking)?	O	10	#20	L2.6
Q29	Is AI controllable, and how do agentic workflows and safety protocols constrain it?	O	9	#09	L3.5

Cluster G — Systems and Measurement (3 modules)

ID	Question	Own	Week	Src	Lec
Q30	How does the PyTorch compiler work and what are its layers? Why does functional autodiff use far more memory than tape-based autodiff?	K	7	#29	L2.9
Q31	Parallelism, GPU memory and power, and the bottlenecks that make AI so expensive	O	7	#13, #15, #31, #36	L1.5
Q32	Can information flow through a model be quantified, for example with entropy?	K	4	#38	L2.1

Cross-review matrix

Each module's author is reviewed by the other two agents:

Author	Modules	Reviewed by
Paradise (11)	Q03, Q09, Q11, Q12, Q13, Q20, Q21, Q22, Q23, Q26, Q27	Ollie + Kukla
Ollie (11)	Q01, Q02, Q04, Q06, Q08, Q16, Q17, Q24, Q28, Q29, Q31	Kukla + Paradise
Kukla (10)	Q05, Q07, Q10, Q14, Q15, Q18, Q19, Q25, Q30, Q32	Paradise + Ollie

Production notes

- **Directory layout:** `modules/Q<NN>-slug/` containing `note.tex` (3–5 pp tutorial), `deck.tex` (Beamer, 5–10 slides, 16:9), `bibliography.bib`, `code/` (runnable, tested), `discussion.tex` (5 questions), `projects.tex` (deeper-project suggestions), `quiz.tex` (comprehension check + answer), and the compiled PDFs.
- **Shared preamble:** one `preamble.sty` at repo root so all 32 modules have a unified look (ink-navy + terracotta palette, PARADISE/OLLIE/KUKLA header signatures for authorship at a glance).
- **Citations:** every bibliography entry verified against arXiv/publisher before merge; no invented references. Anchor papers per cluster come from the master lecture list.
- **Code:** runs on Python 3.10+ with CPU-only defaults; GPU optional. Each module's code has a `README` stating runtime and dependencies.
- **Earliest deadlines** (weeks 4–5 modules): Q01, Q02, Q06, Q20, Q32 (week 4); Q03, Q04, Q05, Q21, Q26 (week 5).